

Mayank Khandelwal

Senior AI/ML Engineer | Pune, India

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Summary

Senior AI/ML Engineer and Data Scientist with **7+ years of experience** building and evaluating production AI systems across **search and query understanding, NLP, Generative AI, retrieval-augmented generation, and computer vision**. Experienced in developing **multi-document research assistants, grounded question-answering workflows, ML evaluation frameworks, low-latency inference pipelines, and scalable FastAPI services**. Strong hands-on background in **Python, PyTorch, TensorFlow, Transformers, vector retrieval, ONNX, Docker, Kubernetes, PostgreSQL, and MLOps**. Proven ability to take ML solutions from problem definition and experimentation through **evaluation, API development, deployment, performance optimisation, monitoring, and production reliability**.

Core Skills

Programming & APIs: Python, SQL, FastAPI, Flask, Uvicorn, Pydantic, REST and asynchronous services.

LLM & Retrieval Systems: RAG, multi-document QA, HyDE, dense retrieval, metadata filtering, citations, grounding, query rewriting, prompt design, LangChain, LangGraph and LlamaIndex.

Evaluation & Observability: Retrieval, grounding and citation evaluation; real and synthetic datasets; human evaluation; RAGAS, Langfuse, LangSmith and tracing.

Machine Learning & NLP: PyTorch, TensorFlow, scikit-learn, Transformers, BERT, sentence-transformers, classification, OOD detection and multi-label NLP.

Computer Vision & Inference: YOLO, OpenCV, detection, segmentation, ONNX, CPU/GPU inference and latency optimisation.

MLOps & Infrastructure: Docker, Kubernetes, MLflow, GitHub Actions, CI/CD, model versioning, monitoring, PostgreSQL, BigQuery, GCP, Vertex AI, Pinecone and FAISS.

Experience

Senior AI/ML Engineer

Oct 2025 – Present

Springer Nature AI Labs

Pune, India

- Core AI/ML engineer on a **researcher-facing beta assistant**, contributing across reading assistance, manuscript advice, related-paper discovery, question generation and multi-paper conversational workflows.
- Built and reviewed an **agentic technical-review workflow** that assesses research papers for technical correctness, clarity and relevance; evaluated quality against peer-review data and human feedback.
- Contributed to multi-paper chat for approximately **20–25 papers**, implementing conversation flow and retrieval integration using **HyDE query rewriting**, dense retrieval with metadata filtering, and citation-backed source attribution.
- Implemented and owned the **multi-paper evaluation framework**, measuring retrieval, grounding, relevance and citation quality across three curated ten-paper collections using real and synthetic questions; findings guided chunking, retrieval and citation-design decisions.
- Owned a separate production **Academic Question Delivery API** end to end: FastAPI/Uvicorn, PostgreSQL, TTL caching, per-key async locking, load testing, Kubernetes, monitoring and reliability. The service returns two pre-generated questions from approximately **1.5M records**, targeting **100–150 RPS** at approximately **600 ms P90**.

Senior Machine Learning Engineer

Jul 2024 – Oct 2025

Tech Mahindra – Client: Samsung Research India

Bangalore, India

- Owned an **out-of-domain query classifier** and its ONNX-based CPU inference pipeline end to end, achieving **>0.90 F1, 0.99 precision** and approximately **10 ms latency** against a 20 ms target.
- Contributed query-understanding signals, embeddings, dense-retrieval components, ranking integration and search analytics for a global e-commerce search platform; did not own the final ranking system.
- The broader search-improvement initiative delivered approximately **35% higher relevance, 25% stronger engagement** and **20% higher conversion**, validated using manually labelled queries and A/B testing.

Senior Machine Learning Engineer

Camcom Technologies

Apr 2023 – Jul 2024

Bangalore, India

- Owned detection and segmentation models and their evaluation for visual-pollution analysis across more than **2 million street-view images**, achieving detection mAP above **0.60** and segmentation mAP@50–95 of approximately **0.27**.
- Built the annotation strategy and contributed to dataset preparation and ONNX export, supporting repeatable training, evaluation and inference workflows.
- The wider production system reduced inference latency by approximately **60%** and manual inspection effort by more than **70%**; GPU optimisation, Triton serving and production deployment were owned by the broader team.

Data Scientist

Tata Consultancy Services

Jun 2019 – Mar 2023

Pune, India

- Built a BERT-based email-classification system with **96% accuracy**, automating classification of more than **500K emails per month** and saving approximately **800 hours** of manual triage.
- Developed Tableau and Power BI dashboards that supported data-driven decisions and reduced operational costs by approximately **15%**; improved Airflow/Spark data pipelines to reduce processing time by approximately **40%**.
- Researched and implemented OpenCV and YOLOv3 computer-vision models, achieving more than **90% precision** for face detection and improving object-classification performance by approximately **15%**.

Certifications

Machine Learning – Andrew Ng, Coursera (Jul 2021) | **TensorFlow Developer Specialization** – DeepLearning.AI (Sep 2020)

Deep Learning Specialization – DeepLearning.AI (Apr 2019) | **MTA: Programming Using Python** – Microsoft (Dec 2020)

Awards & Recognition

Value Creator, Samsung Research India (2025) | **Team Awesome**, Samsung Research India (2025) | **On-the-Spot Best Performer**, TCS (2023)

Selected Publications

Keyword Spotting in Images Using Convolutional Neural Network, IJITEE (2019)

Wearable Technology for Visually Impaired, IJIREM (2018)

Education

Bachelor of Technology in Computer Science, SRM Institute of Science and Technology

2015–2019